SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

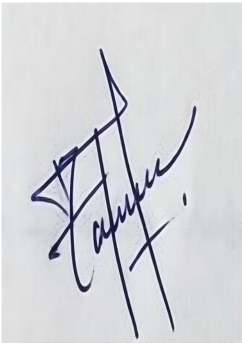
Annexure A

<u>Analytical Problem Solving</u>

Code of Conduct:

I Mohammed Affan M /191572169(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

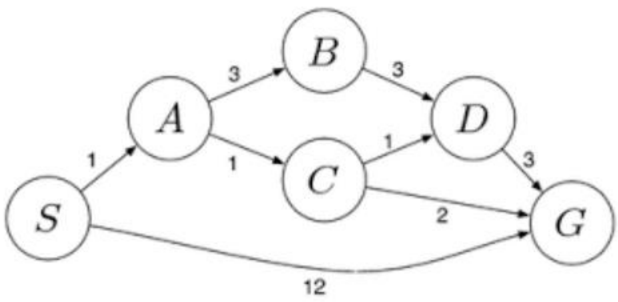
i. What are the percept's for this agent?
ii. Characterize the operating environment.
iii. What are the actions the agent can take?
iv. How can one evaluate the performance of the agent?
v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understandin g	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understandin g	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

		requirements accurately			

CO1-Assesment Tool -1

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ANALYTICAL PROBLEM SOLVING

1.

- Jug A = 4 gallons
- Jug B = 3 gallons

Both jugs are initially empty. The objective is to obtain exactly 2 gallons in the 4-gallon jug.

Represent a state as:

[

(x,y)

]

where (x) is the amount in the 4-gallon jug and (y) is the amount in the 3-gallon jug.

Initial state:

[

(0,0)

]

Goal state:

[

(2,0)

]

Method

The problem can be solved using state-space search. Breadth-First Search is suitable because all actions have equal cost.

Steps

Step	Operation	State
1	Fill the 3-gallon jug	((0, 3))
2	Pour it into the 4-gallon jug	((3, 0))
3	Fill the 3-gallon jug again	((3, 3))
4	Pour into the 4-gallon jug until full	((4, 2))
5	Empty the 4-gallon jug	((0, 2))
6	Pour the remaining 2 gallons into it	((2, 0))

Pseudocode

```
BEGIN
   Fill 3-gallon jug
```

Pour it into 4-gallon jug

Fill 3-gallon jug again

Pour into 4-gallon jug until it is full

Empty 4-gallon jug

Pour remaining water into 4-gallon jug

DISPLAY "2 gallons obtained"

END

Result

The final state is:

[

\boxed{(2,0)}

]

Therefore, exactly 2 gallons are obtained in the 4-gallon jug.

2.

A Mars rover is an autonomous agent used to explore the surface of Mars, analyse rocks and soil, collect scientific data and send the results to Earth.

i. Percepts

The rover receives the following percepts:

- Images of rocks, soil and craters
- Distance from obstacles
- Ground slope and roughness
- Temperature and atmospheric data
- Chemical composition of samples

- Current location and direction
- Battery level
- Wheel and instrument condition

ii. Operating Environment

The Mars rover environment is:

- Partially observable: It cannot see the complete environment.
- Stochastic: Wheel slip or dust may affect actions.
- Sequential: One action affects future actions.
- Dynamic: Temperature, light and dust can change.
- Continuous: Speed, distance and temperature have continuous values.
- Unknown: Terrain conditions may not be known in advance.

iii. Actions

The rover can:

- Move forward or backward
- Turn left or right
- Avoid obstacles
- Capture images
- Drill rocks
- Collect and analyse samples
- Store data
- Send data to Earth
- Enter safe or power-saving mode

iv. Performance Measure

Performance can be evaluated using:

- Number of useful samples collected
- Quality of scientific data
- Safe distance travelled
- Energy consumed
- Obstacles avoided
- Data successfully transmitted
- Mission goals completed

v. Agent Architecture

A hybrid intelligent agent is most suitable. It combines reflex, model-based, goal-based, utility-based and learning behaviour.

Pseudocode

```
WHILE mission is active DO
  Read sensors
  Update internal map

  IF danger is detected THEN
    Avoid danger
  ELSE IF battery is low THEN
    Enter power-saving mode
  ELSE
    Select scientific goal
    Plan safe route
    Execute next action
  END IF
```

Store and transmit data

END WHILE

Result

A hybrid agent is suitable because the rover must react quickly to danger and also plan scientific tasks independently.

3.

Problem

Place eight queens on an (8 \times 8) chessboard so that no queen attacks another queen.

Two queens must not be:

- In the same row
- In the same column
- On the same diagonal

Problem Formulation

- Initial state: Empty chessboard
- State: Positions of queens already placed
- Action: Place a queen in a safe row of the next column
- Goal state: Eight queens placed without conflict
- Path cost: One cost for each queen placement

A state may be represented as:

[

[1,5,8]

]

This means queens are placed in rows 1, 5 and 8 of the first three columns.

Method

Backtracking search is suitable. It places queens column by column. If no safe position is available, it removes the previous queen and tries another position.

A queen at ((r,c)) is unsafe when:

[

$r = previousRow$

]

or

[

$|r - previousRow| = |c - previousColumn|$

]

Pseudocode

```
FUNCTION PlaceQueen(column)

   IF column > 8 THEN
      RETURN TRUE
   END IF

   FOR row     1 TO 8 DO

      IF position is safe THEN
         Place queen at row, column
```

```
      IF PlaceQueen(column + 1) THEN
        RETURN TRUE
      END IF

      Remove queen
    END IF

  END FOR

  RETURN FALSE
END FUNCTION
```

One Valid Solution

$$
\boxed{[1,5,8,6,3,7,2,4]}
$$

This means one queen is placed in every column using the above row positions.

Result

Backtracking is efficient because it rejects invalid arrangements immediately instead of checking every possible board configuration.

4.

Problem

A customer wants to travel from a pickup location to a destination using the OLA application. The customer may choose Micro, Mini, Sedan, Shared or Prime based on fare, time and comfort.

Type of Agent

A goal-based problem-solving agent with utility-based selection is suitable.

The main goal is:

[

\text{Reach the destination}

]

The utility component selects the best cab using fare, waiting time, travel time and comfort.

Problem Formulation

- Initial state: Pickup location, destination and available cabs
- Actions: Search cabs, calculate route, estimate fare and book driver
- Goal state: Customer reaches the destination
- Path cost: Fare, travel time, waiting time and distance

A simple utility function is:

[

Utility = Comfort-Fare-WaitingTime-TravelTime

]

A* search can be used to find a suitable route.

[

f(n)=g(n)+h(n)

]

where (g(n)) is the actual cost and (h(n)) is the estimated remaining cost.

Pseudocode

```
FUNCTION BookCab(pickup, destination)

  cabs    FindNearbyCabs(pickup)

  IF no cab is available THEN
    RETURN "No cab available"
  END IF

  route    FindBestRoute(pickup, destination)

  FOR each cab DO
    fare    EstimateFare(cab, route)
    wait    EstimateWaitingTime(cab)
    utility    comfort - fare - wait
  END FOR

  Display ranked cab options
  selectedCab    CustomerSelection()
  RequestDriver(selectedCab)
```

```
    IF driver accepts THEN
        ConfirmBooking
        Track cab until destination
    ELSE
        Request next driver
    END IF

END FUNCTION
```

Result

The goal-based agent finds a way to reach the destination, while the utility-based part selects the most suitable cab.

5.

Problem

A logistics company wants to transport a package from warehouse (S) to warehouse (G) at minimum cost. Each route has a travel cost representing fuel and time.

Uniform Cost Search is suitable because it always expands the path having the lowest total cost.

Assumed Graph

Since the graph is not included in the given text, consider the following example:

Route	Cost
S to A	2

S to B	5
A to C	2
C to G	6
B to D	1
D to G	2

UCS Process

Start with:

[

$S(0)$

]

1. Expand (S): add (A(2)), (B(5))
2. Expand (A(2)): add (C(4))
3. Expand (C(4)): add (G(10))
4. Expand (B(5)): add (D(6))
5. Expand (D(6)): update (G) to cost (8)
6. Expand (G(8)): goal reached

The least-cost path is:

[

$S\rightarrow B\rightarrow D\rightarrow G$

]

Total cost:

[

5+1+2=8

]

Pseudocode

```
FUNCTION UCS(start, goal)

  Insert start into priority queue with cost 0

  WHILE queue is not empty DO

    Remove node with lowest cost

    IF node = goal THEN
      RETURN path and cost
    END IF

    FOR each neighbour DO
      newCost    currentCost + edgeCost

      IF newCost is lower THEN
        Update neighbour cost
        Store parent
```

```
        Insert into queue
      END IF
    END FOR

  END WHILE

END FUNCTION
```

Result

The minimum-cost route in the assumed graph is:

[

\boxed{S\rightarrow B\rightarrow D\rightarrow G}

]

with a total cost of:

[

\boxed{8}

]

Uniform Cost Search gives the optimal path when all route costs are positive.